

Economics 2220

Principles Of Macroeconomics

Professor Zhigang Feng

Lecture 12:

Fiscal Policy

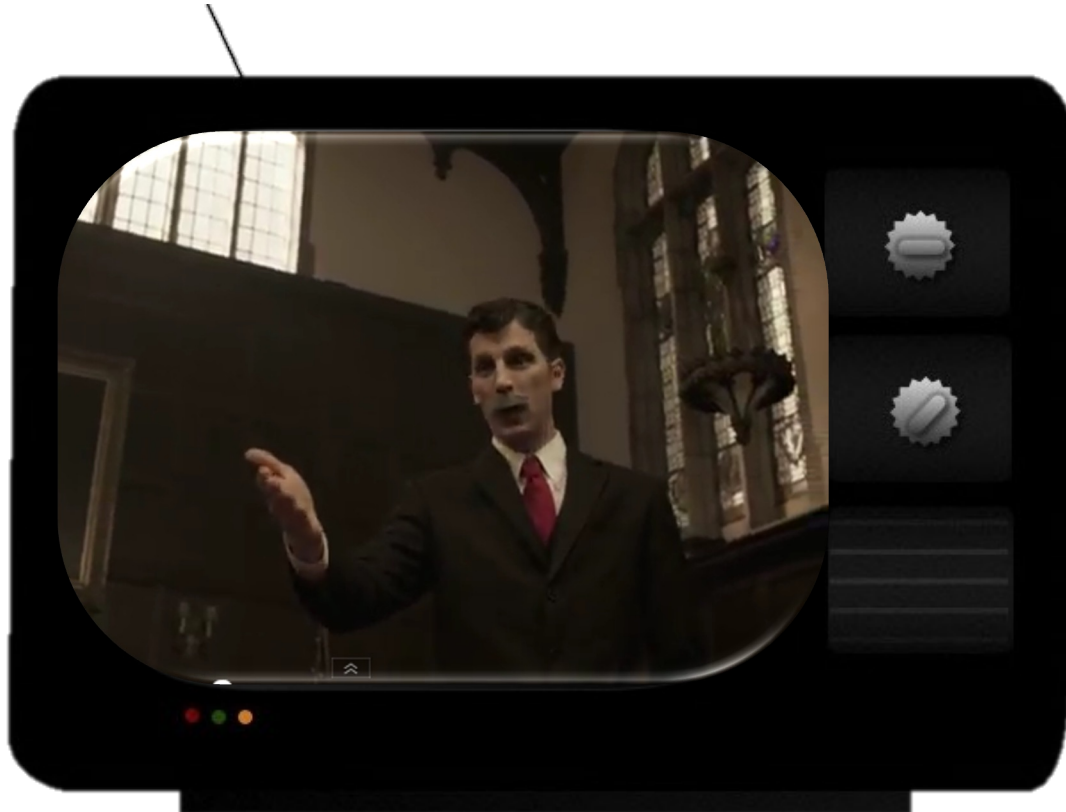
► What you will learn in this chapter



- What fiscal policy is and why it is an important tool in managing economic fluctuations
- Which policies constitute an expansionary fiscal policy and which constitute a contractionary fiscal policy
- Why fiscal policy has a multiplier effect and how this effect is influenced by automatic stabilizers
- Why governments calculate the cyclically adjusted budget balance
- Why a large public debt may be a cause for concern
- Why implicit liabilities of the government are also a cause for concern

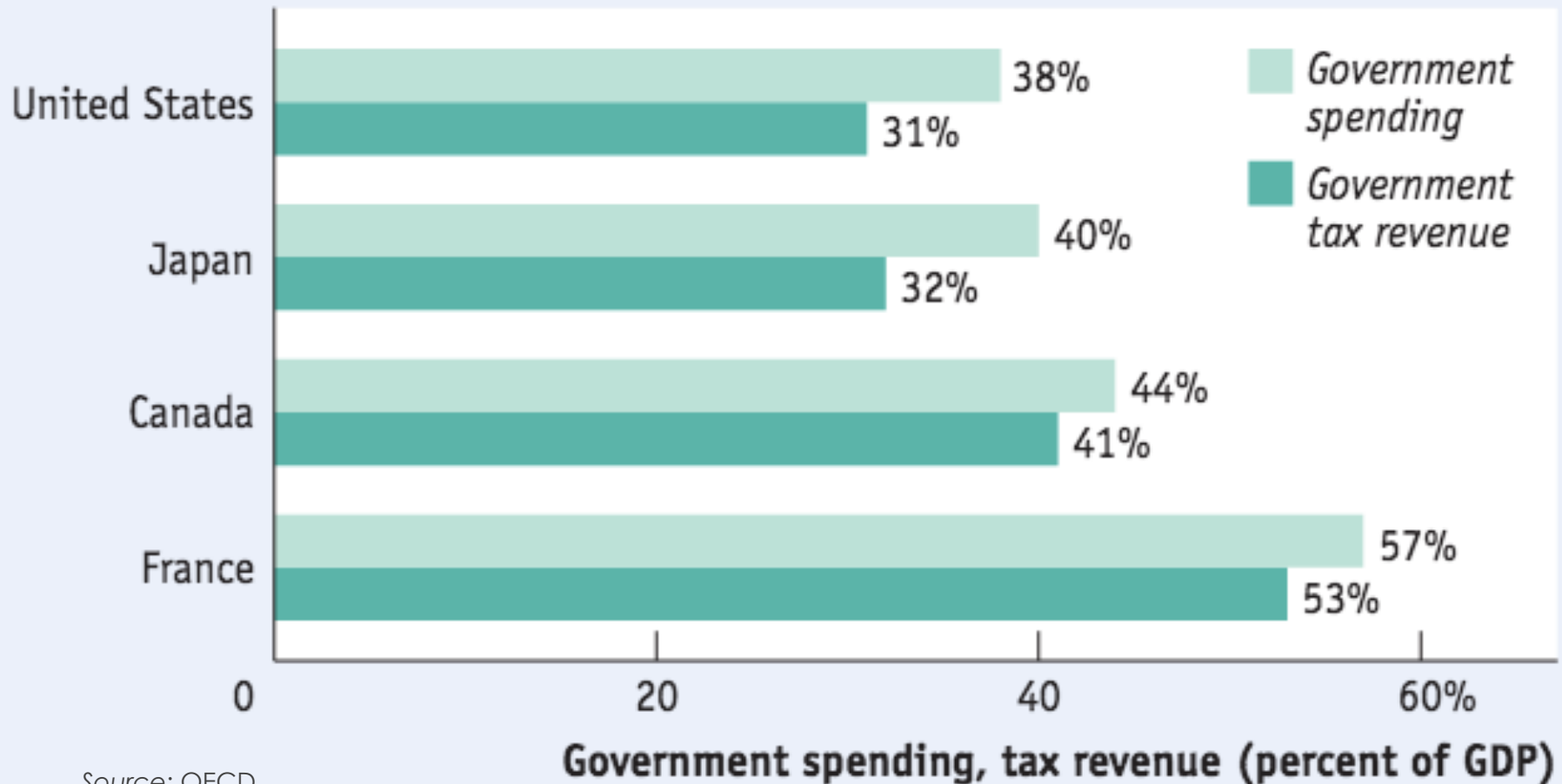
LEARN BY DOING: APPLICATION VIDEO

They're back! Keynes vs. Hayek: the fight of the century, round 2. Is fiscal policy stabilizing or destabilizing? Has the Great Recession proved either man right? Click [here](#) or below. (10:10 minutes)



FISCAL POLICY: THE BASICS

Government spending and tax revenue for some high-income countries in 2013



FISCAL POLICY: THE BASICS

The government funds many programs through tax revenues.

Some important terms:

Government transfers: payments by the government to households for which no good or service is provided in return.

Social insurance programs: government programs (transfer payments) intended to protect families against economic hardship.

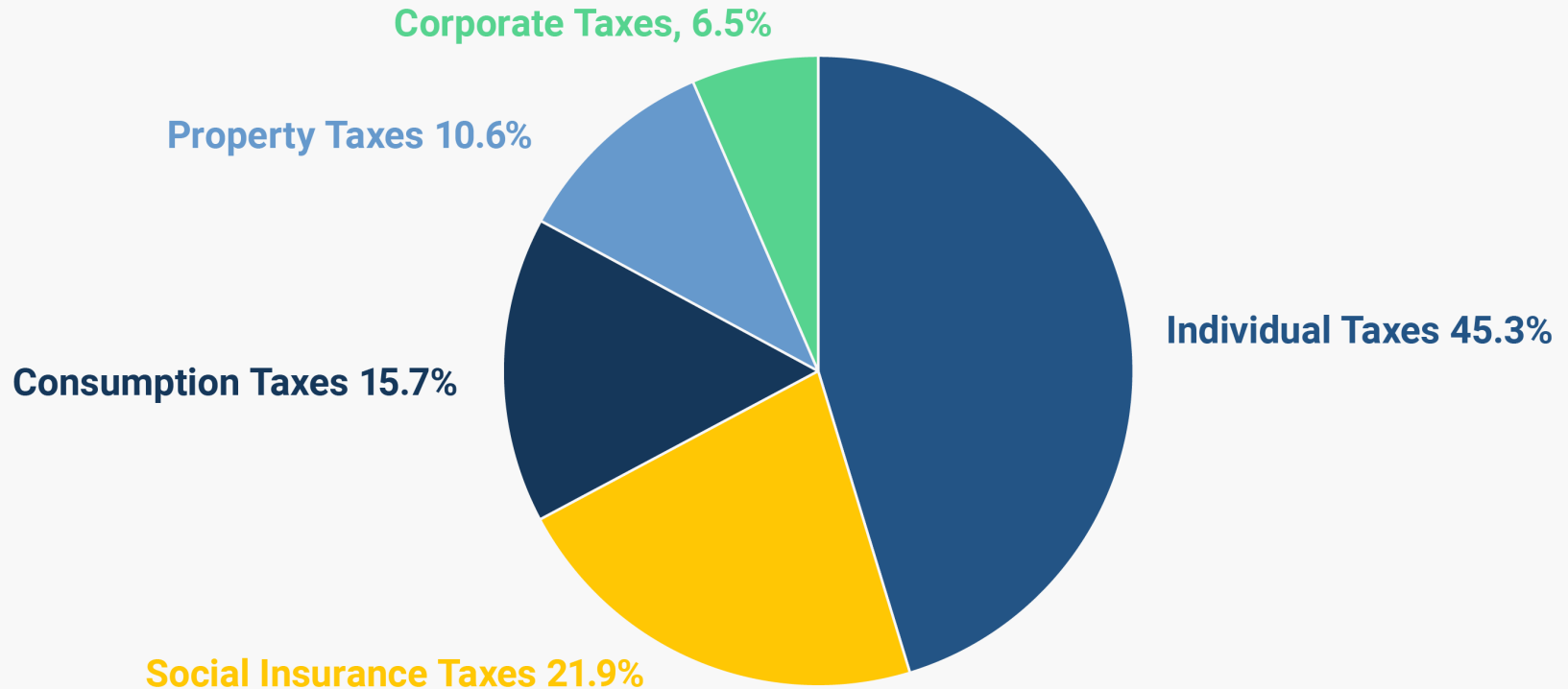
Social Security

Medicare

Medicaid

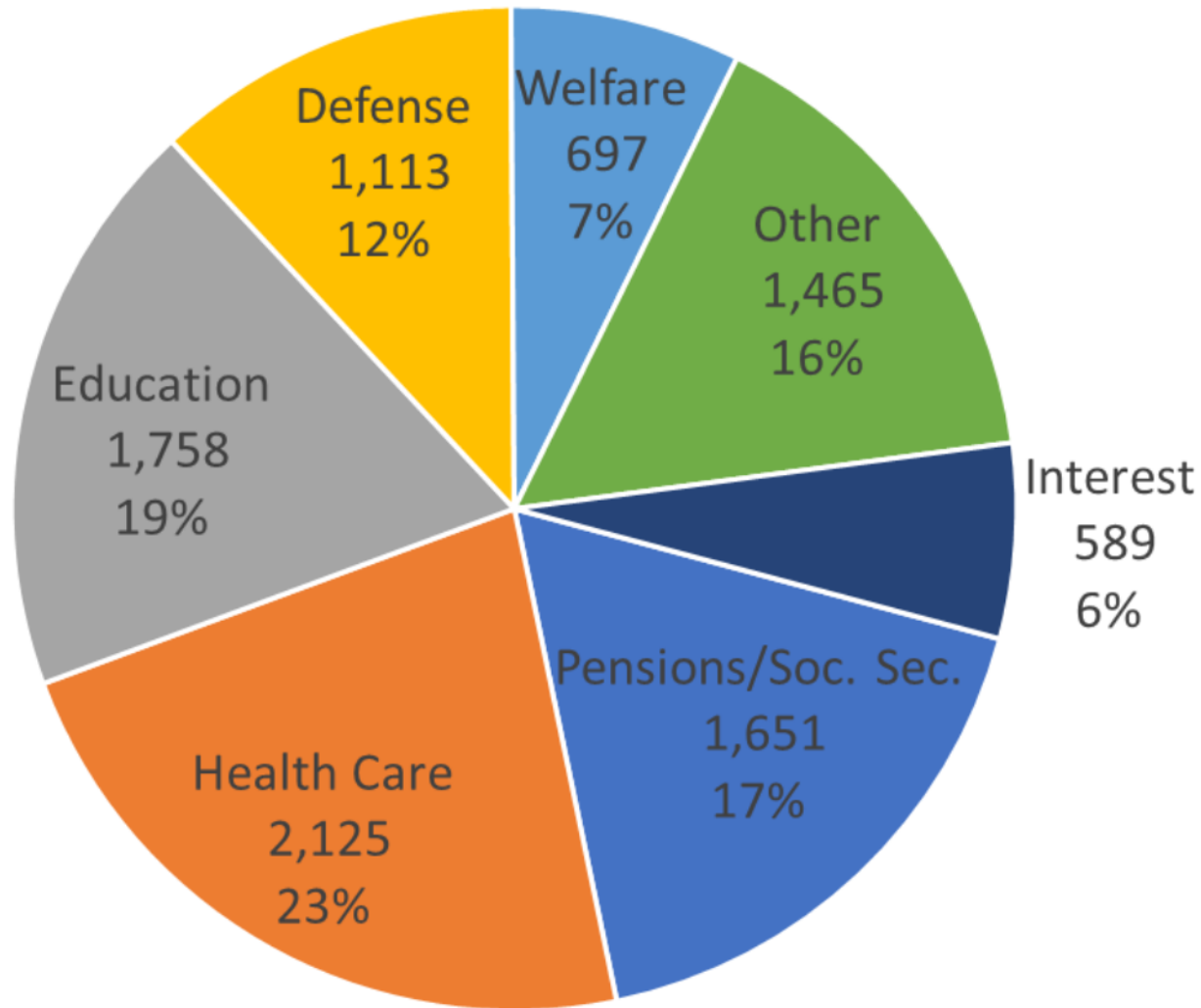
Individual Taxes Are the Most Important Tax Revenue Source for the United States

Sources of Tax Revenue in the United States, 2022



Source: OECD, "Revenue Statistics - OECD Countries Comparative Tables"

GOVERNMENT SPENDING IN THE U.S., 2023



Source: Bureau of Economic Analysis

LEARN BY DOING: PRACTICE QUESTION

Government programs that are designed to protect individuals or families from economic hardship are described as:

- a) fiscal policy.**
- b) discretionary fiscal policy.**
- c) monetary policy.**
- d) social insurance.**

THE GOVERNMENT BUDGET AND TOTAL SPENDING

$$GDP = C + I + G + X - IM$$

The government directly controls **G** and indirectly affects **C** and **I**.

How?

Household incomes are affected by taxes and transfers, and business investment is affected by taxes and regulations.

So the government can shift the AD curve.

FISCAL POLICY

Fiscal policy: the use of taxes, government transfers, or government purchases of goods and services to shift the aggregate demand curve.

I tried a tax cut.



Donald, I'm gonna go in a different direction.

EXPANSIONARY AND CONTRACTIONARY FISCAL POLICY

Expansionary fiscal policy: fiscal policy that increases aggregate demand:

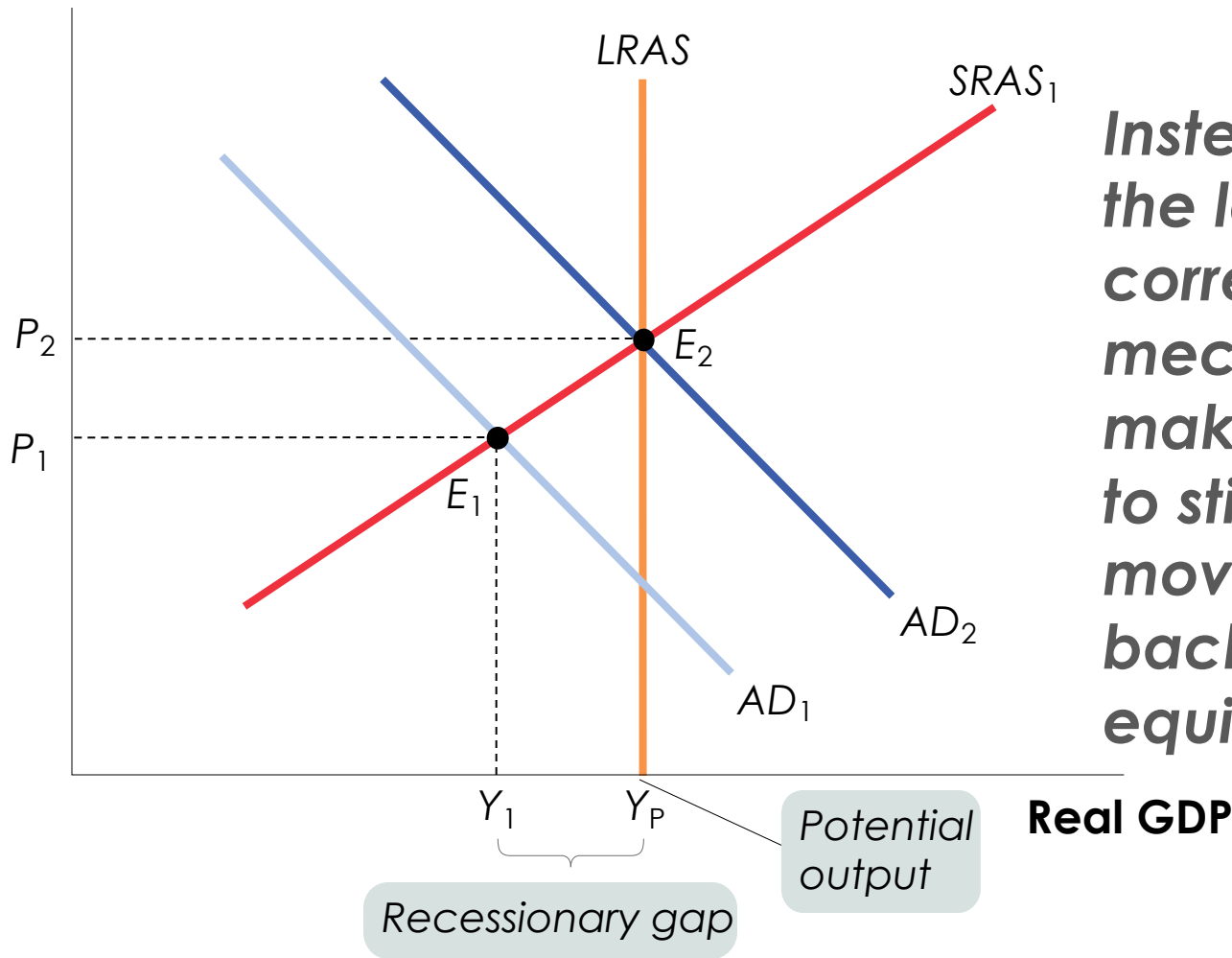
- *an increase in government purchases of goods and services*
- *a cut in taxes*
- *an increase in government transfers*

***Expansionary fiscal
policy:
extra fuel for the
economy***



EXPANSIONARY FISCAL POLICY CAN CLOSE A RECESSIONARY GAP

Aggregate
price level



Instead of waiting for the long-run correction mechanism, policy makers could choose to stimulate AD and move the economy back toward long-run equilibrium.

EXPANSIONARY AND CONTRACTIONARY FISCAL POLICY

Contractionary fiscal policy: fiscal policy that decreases aggregate demand:

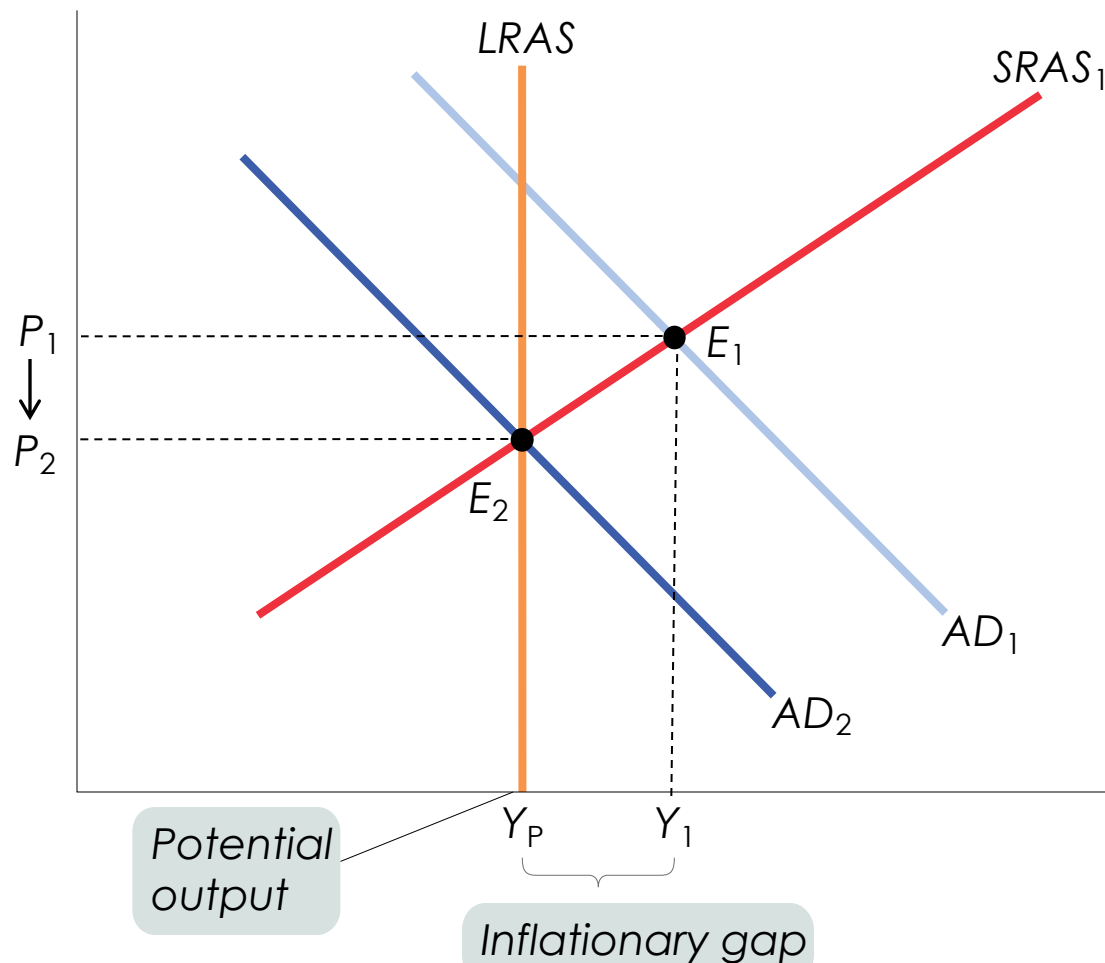
- *a reduction in government purchases of goods and services*
- *an increase in taxes*
- *a reduction in government transfers*

***Contractionary fiscal
policy:
brakes for the economy***



CONTRACTIONARY FISCAL POLICY CAN CLOSE AN INFLATIONARY GAP

Aggregate
price level



Instead of waiting for the long-run correction mechanism, policy makers could choose to contract AD and move the economy back toward long-run equilibrium.

Real GDP

LEARN BY DOING: PRACTICE QUESTION



Do you think the government is right to begin massive spending programs during deep recessions?

- a) Yes
- b) No

LEARN BY DOING: PRACTICE QUESTION

Suppose a country wishes to produce at its potential output level. Holding everything else constant, identify which of the following policy initiatives might help it reach this goal and how these policy initiatives would help.

- a) The government initiates policies that encourage private investment spending.**
- b) The government increases the amount of money it borrows in the loanable funds market to increase its level of government spending in the economy.**
- c) The government increases taxes on consumers and corporations.**
- d) The government authorizes new spending programs.**

LEARN BY DOING: PRACTICE QUESTION

(answers to previous question)

Any policy initiative that shifts *AD* to the right will help the country move toward its potential output level. Items (a) and (d) will both shift *AD* to the right: government policies that stimulate private investment spending lead to higher levels of aggregate spending and a rightward shift in *AD*; and new spending by government will also lead to higher levels of aggregate spending and a rightward shift in *AD*. Item (b) will cause the *AD* to shift to the right unless the increase in government borrowing crowds out private investment to the point where private investment decreases by an amount equal to the increase in government spending. In that case item (b) would result in no shift in the *AD* curve. Item (c) will reduce disposable income and lead to lower levels of aggregate spending: *AD* may shift to the left if the decrease in spending is greater than the increase in government spending, or *AD* may shift to the right if the increase in government spending is greater than the decrease in spending

CAN EXPANSIONARY FISCAL POLICY ACTUALLY WORK?

There are critics who argue:

**“Government
spending always
crowds out private
spending.”**

**“Government
borrowing always
crowds out private
investment spending.”**

**“Government
budget deficits
reduce private
spending.”**

CAN EXPANSIONARY FISCAL POLICY ACTUALLY WORK?

Claim 1: “Government spending always crowds out private spending.”

The statement is wrong because it **assumes a zero-sum game** in which the **aggregate income** earned in the economy **is always a fixed sum**—which isn’t true. It also **assumes that resources in the economy are always fully employed**—and *the only way to increase government spending is at firms’ expense.*



You're squeezed only if the hot tub is full.

CAN EXPANSIONARY FISCAL POLICY ACTUALLY WORK?

Claim 2: “Government borrowing always crowds out private investment spending.”

This is true only part of the time: **It depends upon whether the economy is depressed.**

If it is, a *fiscal expansion will lead to higher incomes*, which *lead to increased savings*.

The CARES Act of 2020 was a case in point: **despite high levels of government borrowing, U.S. interest rates stayed near historic lows.**



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CAN EXPANSIONARY FISCAL POLICY ACTUALLY WORK?

Claim 3: “Government budget deficits reduce private spending.”

This is known as “*Ricardian equivalence*” (after the nineteenth-century economist David Ricardo).

It assumes that consumers, **seeing the higher debt levels**, will **cut their spending today to save for inevitable increases in future tax rates necessary to pay down the debt**.

Does this give **too much credit to consumers’ foresight and budgeting discipline?** Probably.

A CAUTIONARY NOTE: LAGS IN FISCAL POLICY

In the case of fiscal policy, there is an important reason for caution: There are significant *lags* in its use.

It takes time to:

1. *realize the recessionary or inflationary gap by collecting and analyzing economic data.*
2. *develop a plan.*
3. *implement the action plan (spending the money).*



LEARN BY DOING: PRACTICE QUESTION

Contractionary fiscal policy:

- a) is most helpful for restoring an economy to the potential output level of production when there is a recessionary gap.
- b) shifts the *AD* curve to the right, restoring the equilibrium level of output to the potential output level for the economy.
- c) often causes inflation or an increase in the aggregate price level.
- d) if effective, shifts *AD* to the left, resulting in a reduction in the aggregate output and the aggregate price level for a given short-run aggregate supply curve (*SRAS*).

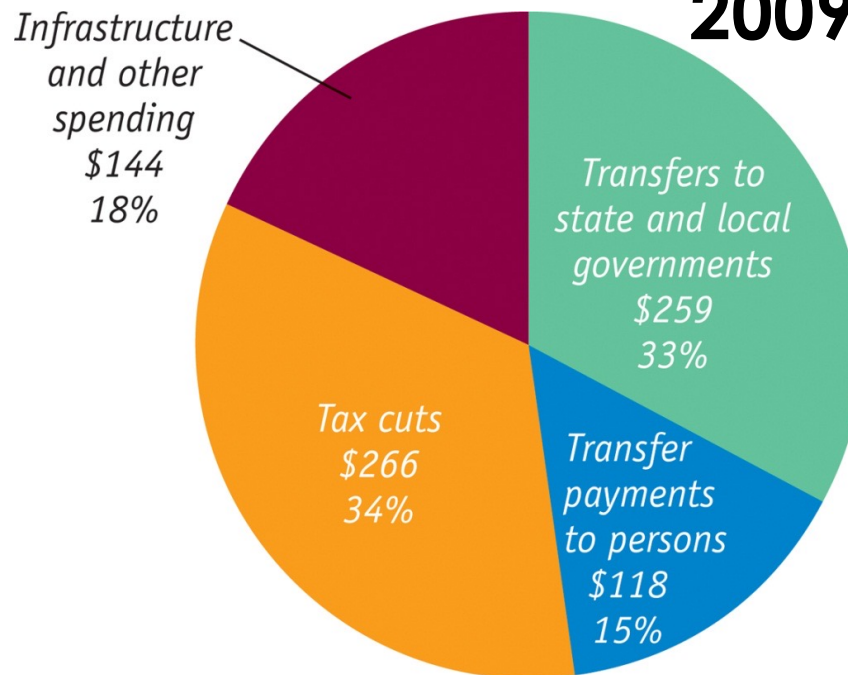
LEARN BY DOING: PRACTICE QUESTION

Which of the following statements is true? Holding everything else constant:

- a) an economy can eliminate an inflationary gap by increasing government spending.**
- b) expansionary fiscal policy refers to an increase in taxes.**
- c) when potential output is greater than actual aggregate output, the economy faces a recessionary gap.**
- d) when *SRAS* intersects *AD* to the right of the long-run aggregate supply (*LRAS*) curve, the economy faces a recessionary gap.**

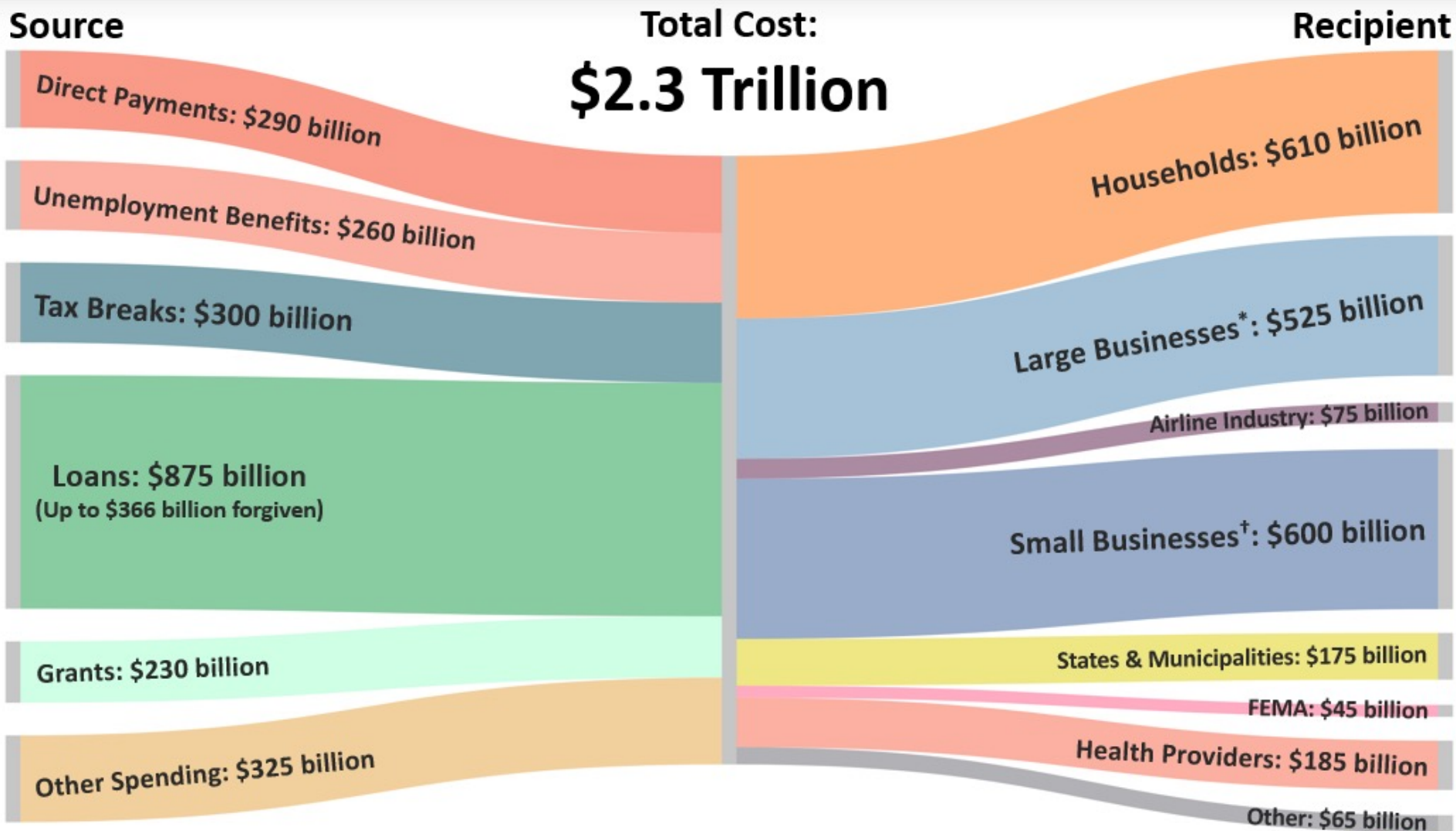
WHAT WAS IN THE RECOVERY ACT?

The American Recovery and Reinvestment Act of 2009 (billions of dollars)



One effect of the recession was a sharp drop in revenues at the state and local levels, which in turn forced these lower levels of government to cut spending. Federal aid was sent to mitigate these cuts.

A Visualization of the CARES Act



† This includes \$170 billion of tax cuts for businesses other than corporations, some of which are large companies.

* This includes \$454 billion to set up a \$4.5 trillion Fed facility, which could help support large businesses but also small businesses and state and local governments.

Source: Legislative offices, JCT, bill text, CRFB estimates.

CRFB.org



THE MULTIPLIER



FISCAL POLICY AND THE MULTIPLIER

Multiplier effects of an increase in government purchases of goods and services:

Recall that (if we assume a simple case with no taxes or international trade),

the multiplier is $1/(1 - MPC)$

Example: if $MPC = 0.5$, the multiplier would be $1/(1 - 0.5) = 2$. So \$50 billion of new government spending would create $(\$50 \text{ billion}) \times 2 = \100 billion increase in real GDP.

FISCAL POLICY AND THE MULTIPLIER

Multiplier effects of changes in government transfers and taxes

Will a \$50 billion tax cut (or increase in transfers) have the same effect as a \$50 billion increase in government purchases?

No. Example: if the $MPS=0.5$, a change in tax or transfers is smaller than an equivalent change in government purchases from the outset.

Effect on real GDP	\$50 billion rise in government purchases of goods and services	\$50 billion rise in government transfer payments
First round	\$50 billion	\$25 billion
Second round	\$25 billion	\$12.5 billion
Third round	\$12.5 billion	\$6.25 billion
•	•	•
•	•	•
•	•	•
Eventual effect	\$100 billion	\$50 billion

FISCAL POLICY AND THE MULTIPLIER

The **size of the shift of the aggregate demand curve** depends on the type of fiscal policy.

Changes in government purchases have a more powerful effect on the economy than equal-sized changes in taxes or transfers.



FISCAL POLICY AND THE MULTIPLIER

A few notes:

It's more complicated, because (unlike most *real* tax policy) we use simple *lump-sum taxes*: taxes that *don't depend* on the taxpayer's income.

If it's not a lump-sum tax, the tax revenue will depend on the level of real GDP (and reduce the size of the multiplier).

TYPES OF FISCAL POLICY

Automatic stabilizers: government spending and taxation rules that cause fiscal policy to be **automatically expansionary when the economy contracts** and **automatically contractionary when the economy expands** (*unemployment insurance*).

In contrast, ***discretionary fiscal policy*** arises from **deliberate actions** by policy makers rather than rules (*the Obama stimulus*).



LEARN BY DOING: PRACTICE QUESTION

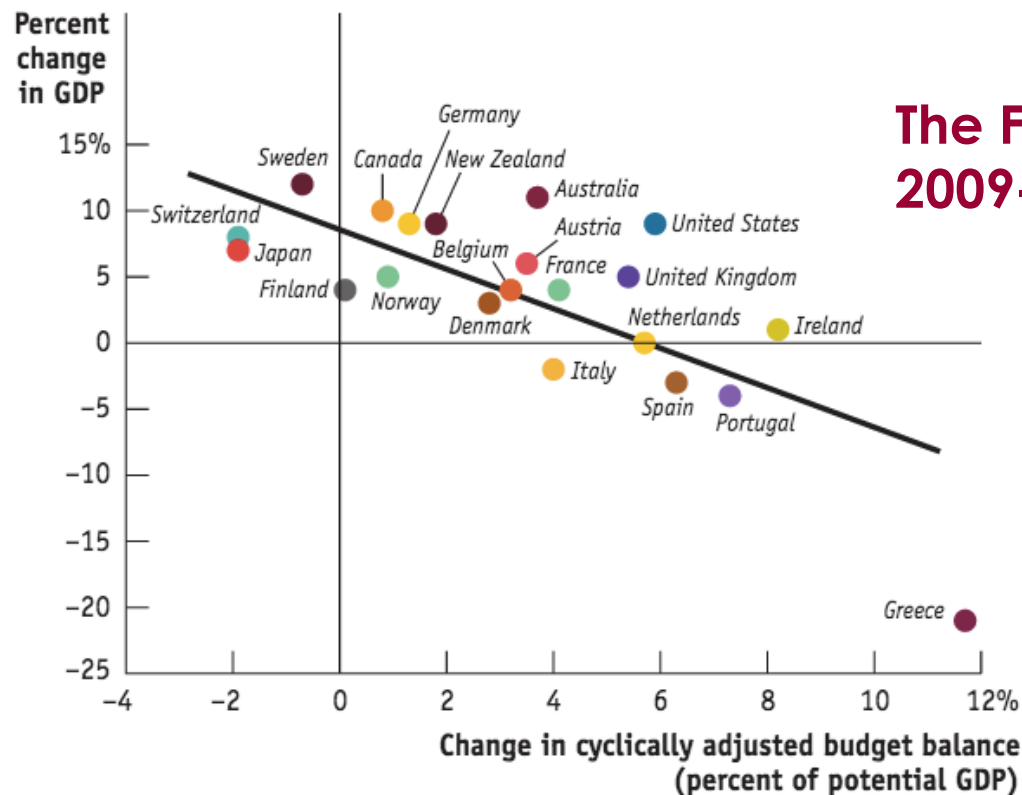
Holding everything else constant, the multiplier effect for taxes or transfers:

- a) is the same as the multiplier effect for changes in autonomous aggregate spending.**
- b) is smaller than the multiplier effect for changes in autonomous aggregate spending.**
- c) is larger than the multiplier effect for changes in autonomous aggregate spending.**
- d) may be smaller than, larger than, or equal to the multiplier effect for changes in autonomous aggregate spending.**

Austerity and the Multiplier

New evidence from Europe suggests that the “multiplier effect” is real...

Several nations experimented with “austerity” (sharp cuts in spending plus tax increases) because of debt concerns... but many did not.



**The Fiscal Multiplier,
2009-2013**

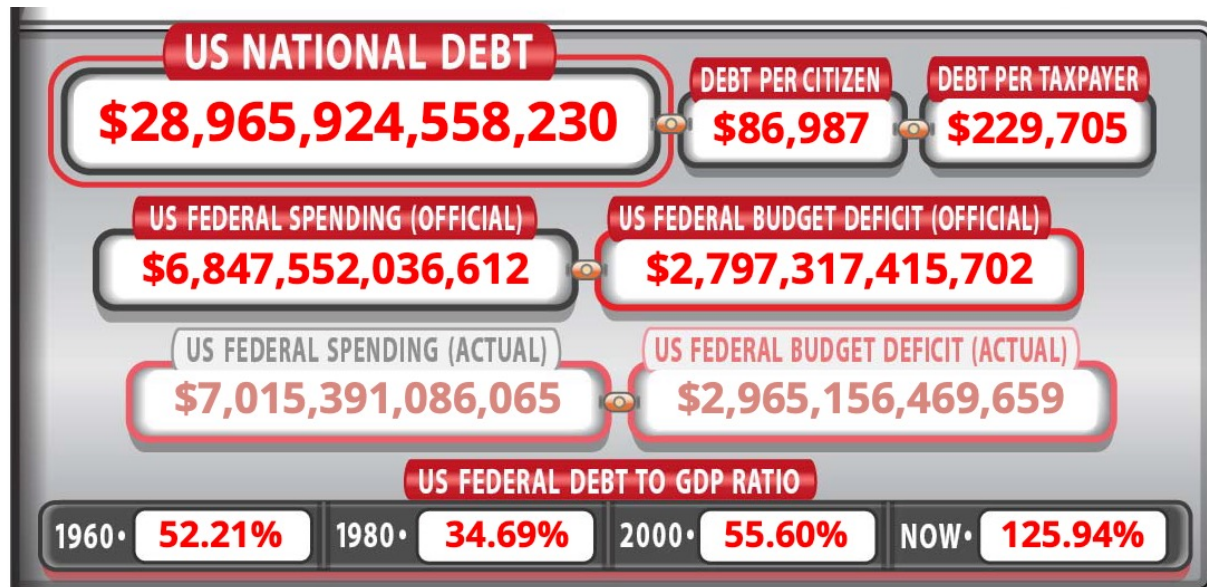
Sources: OECD; World Development Indicators.

THE BUDGET BALANCE

How do surpluses and deficits fit into the analysis of fiscal policy?

Are deficits ever a good thing and surpluses a bad thing?

(And what's the difference between deficit and debt?)



Click [here](#) for real-time debt clock

THE BUDGET BALANCE MEASURES FISCAL POLICY

$$S_{\text{Government}} = T - G - TR$$

Government saving (**S**urplus) = tax revenues (**T**) – government purchases (**G**) and transfers (**TR**).

A **budget surplus** is a positive budget balance, and a **budget deficit** is a negative budget balance.

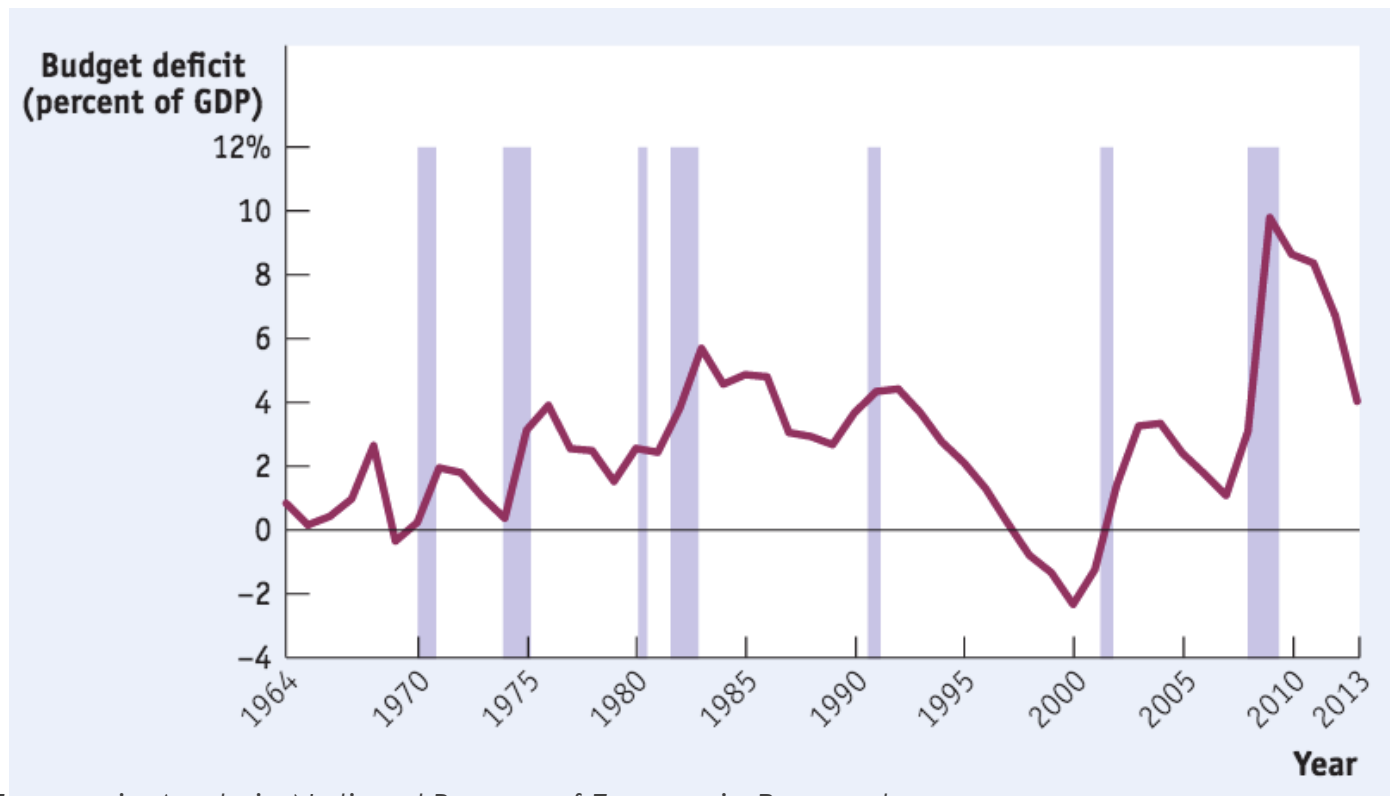
*Other things equal, discretionary **expansionary fiscal policies** reduce the budget balance for that year.*

*Other things equal, discretionary **contractionary fiscal policies** increase the budget balance for that year.*

THE BUDGET BALANCE AND THE BUSINESS CYCLE

Some of the fluctuations in the budget balance are due to the effects of the business cycle.

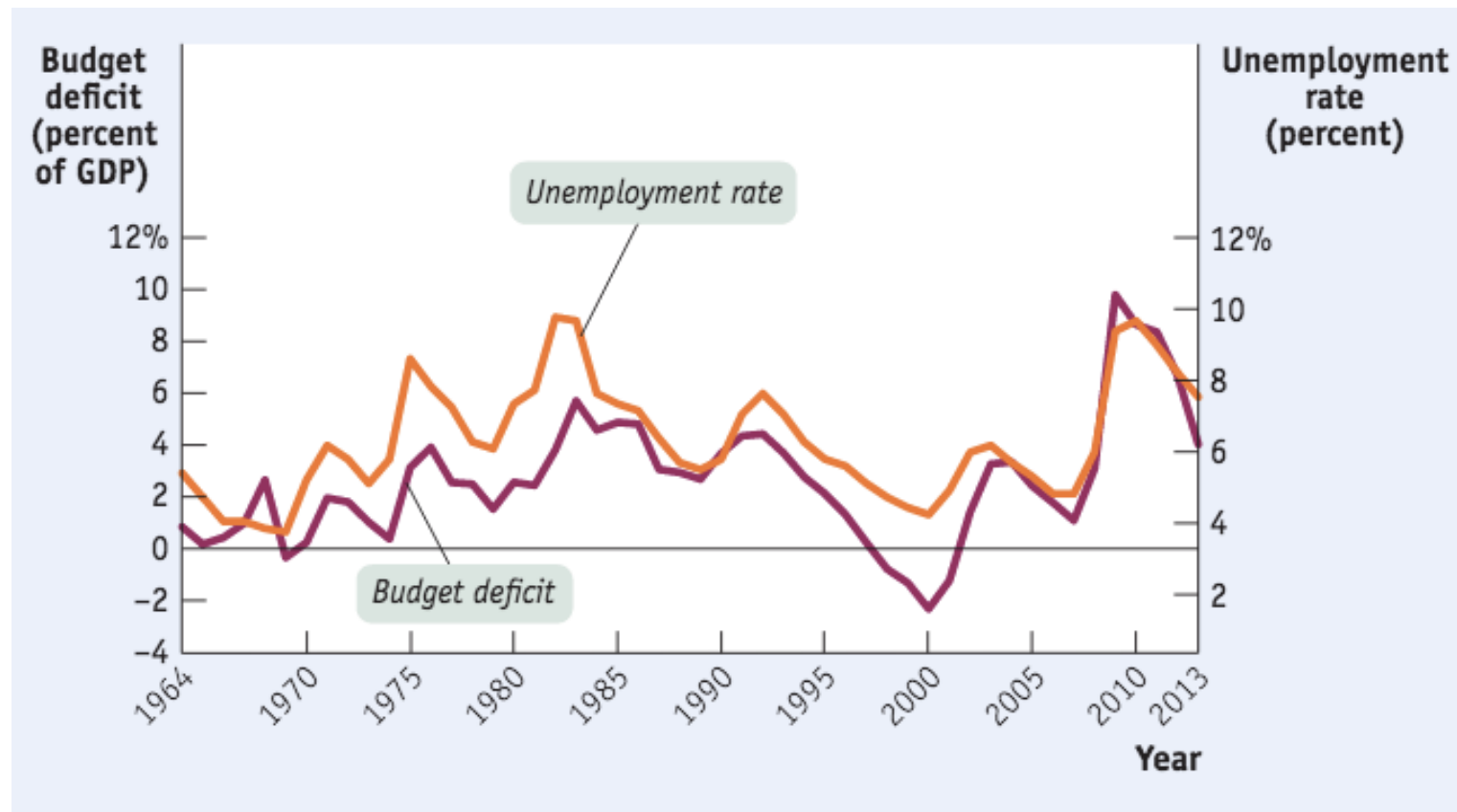
(Shaded areas are recessions.)



Sources: Bureau of Economic Analysis; National Bureau of Economic Research.

THE BUDGET BALANCE AND THE BUSINESS CYCLE

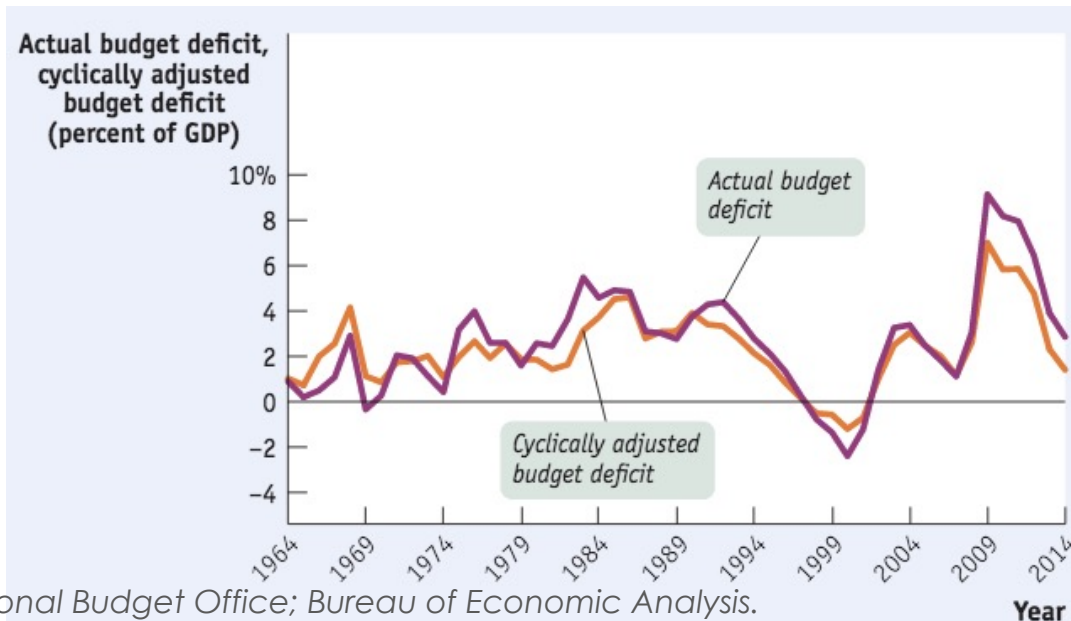
The budget deficit as a percentage of GDP moves closely with the unemployment rate.



Sources: Bureau of Economic Analysis; Bureau of Labor Statistics

THE BUSINESS CYCLE AND THE CYCLICALLY ADJUSTED BUDGET BALANCE

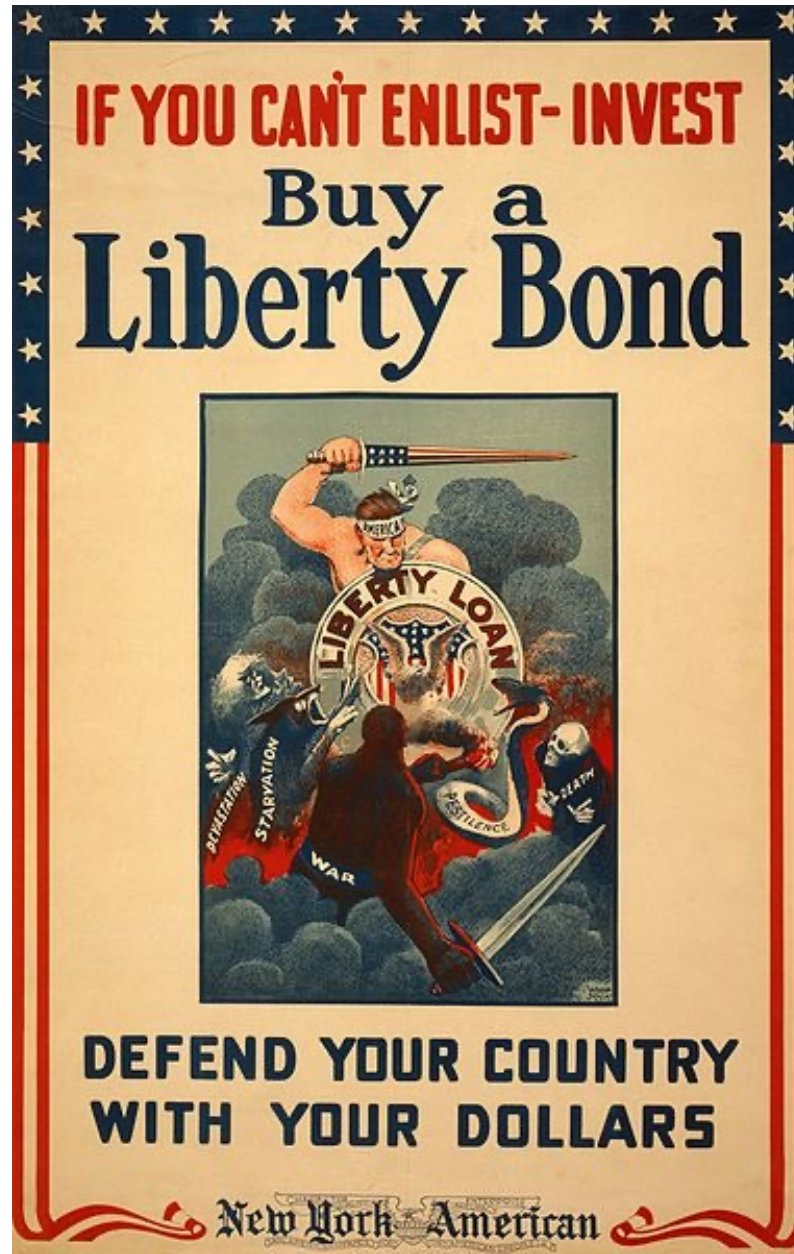
To **separate the effects of the business cycle** from the effects of discretionary fiscal policy, governments estimate the **cyclically adjusted budget balance**: an estimate of the budget balance if the economy were at potential output.



Years of large budget deficits also tend to be years when the economy has a large recessionary gap.

WORLD WAR I WAR BOND

The U.S. deficit rises (and so does the debt) during war times.



LEARN BY DOING: PRACTICE QUESTION

Holding everything else constant, the government's deficit:

- a) tends to increase during a recession.**
- b) tends to increase during an expansion.**
- c) will increase if the government pursues expansionary fiscal policy.**
- d) Answers (a) and (c) are both correct.**
- e) Answers (b) and (c) are both correct.**

PHILOSOPHIES ON BALANCING THE BUDGET

There are various philosophies about requiring a balanced budget:



Require an annually balanced budget...

...and lose the ability to help during a recession.

Balance the budget over the business cycle on average...

...and trust the politicians to keep the long-run budget healthy?

ECONOMICS IN ACTION

EUROPE'S SEARCH FOR A FISCAL RULE

1999: some European nations adopted a common currency, the euro.

Each government was required to keep its budget deficit below 3% of the country's GDP or face penalties.

This pact limited a country's ability to use fiscal policy.

Then, after the 2008 financial crisis, Greece, Ireland, Portugal, Spain, and Italy lost the confidence of their investors, who worried they couldn't pay their debt.

Was it too much debt? No, only Greece had a deficit above 3% of GDP; Ireland and Spain ran surpluses.

BUDGET POLICY IN THE UNITED STATES

The United States has its own version of the original, flawed European stability pact.

The constitutions of 49 of the 50 states require a balanced budget every year.

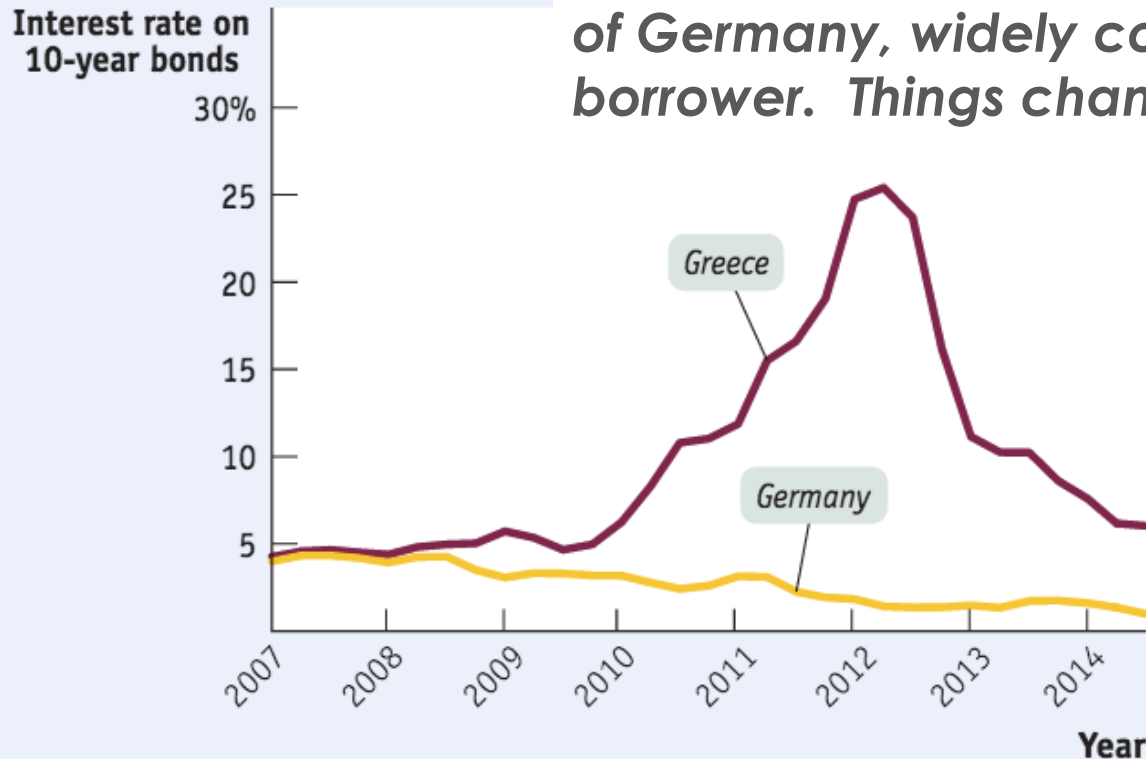
When recession struck in 2008, most states were forced to slash spending and raise taxes in the face of a recession, exactly the wrong thing from a macroeconomic point of view.

Europe has since changed to a rule that requires a mostly balanced structural budget.

LONG-RUN IMPLICATIONS OF FISCAL POLICY

Persistent budget deficits have long-run consequences because they lead to an increase in public debt.

In 2008, the government of Greece could borrow at interest rates barely higher than those of Germany, widely considered a very safe borrower. Things changed in 2009...



Source: Federal Reserve Bank of St. Louis; OECD "Main Economic Indicators Complete Database."

THE SOCIAL COST OF DEBT



Greece imposed severe spending cuts to qualify for loans. Greeks angered by their government's harsh austerity measures took to the streets in protest- and installed a new party to power.

DEFICITS VS. DEBT

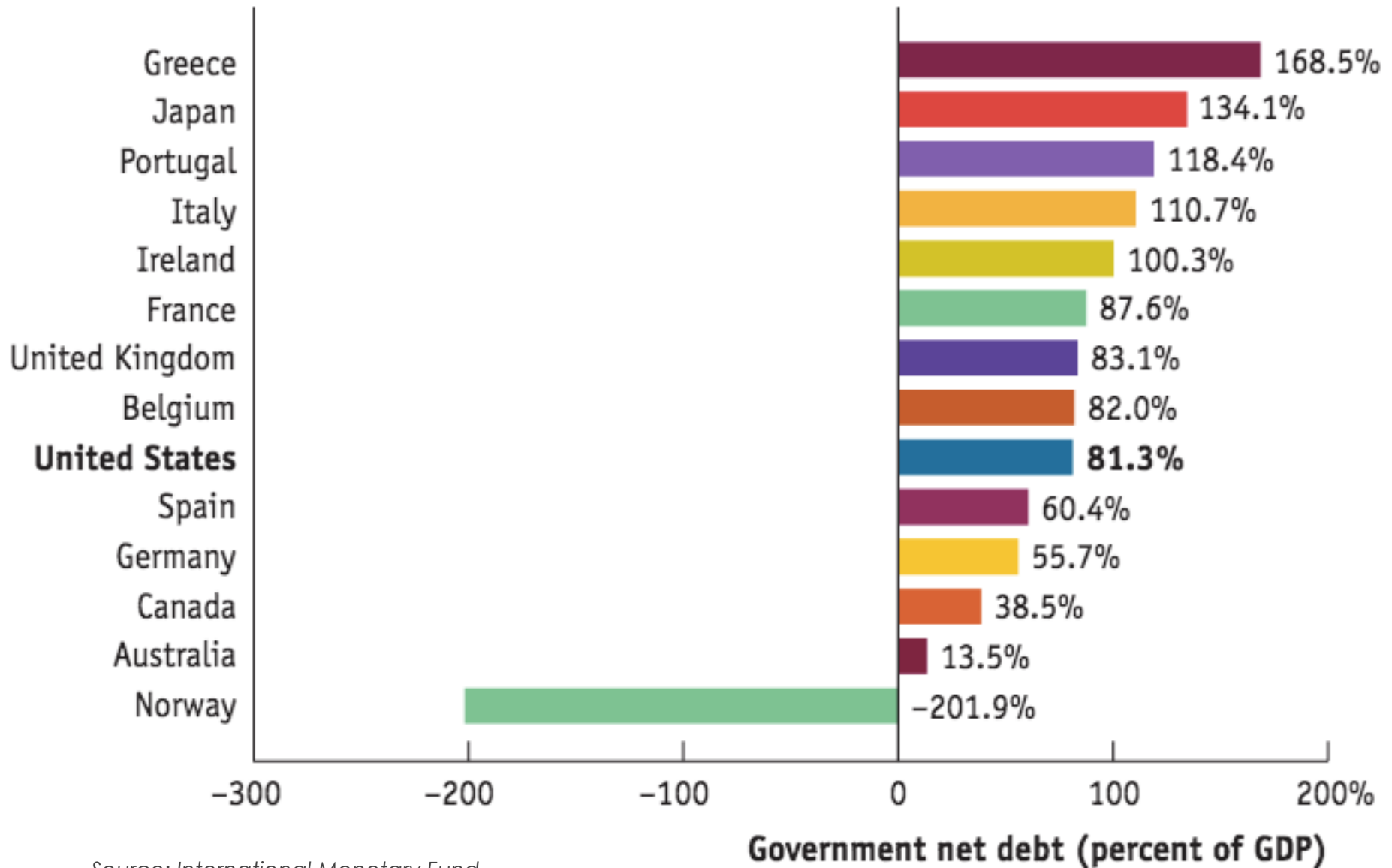
A **deficit** is the difference between the amount of money a government spends and the amount it receives in taxes **over a given period**.

A **debt** is the **sum** of money a government owes **at a particular time**.

Deficits and debt are linked, because government debt grows when governments run deficits. But they aren't the same thing, and they can tell different stories.

Italy found itself in debt trouble in 2011. It had a fairly small deficit by historical standards, but it had very high debt, a legacy of past policies.

THE AMERICAN WAY OF DEBT



Source: International Monetary Fund

PROBLEMS POSED BY RISING GOVERNMENT DEBT

Public debt may crowd out investment spending, which reduces long-run economic growth.

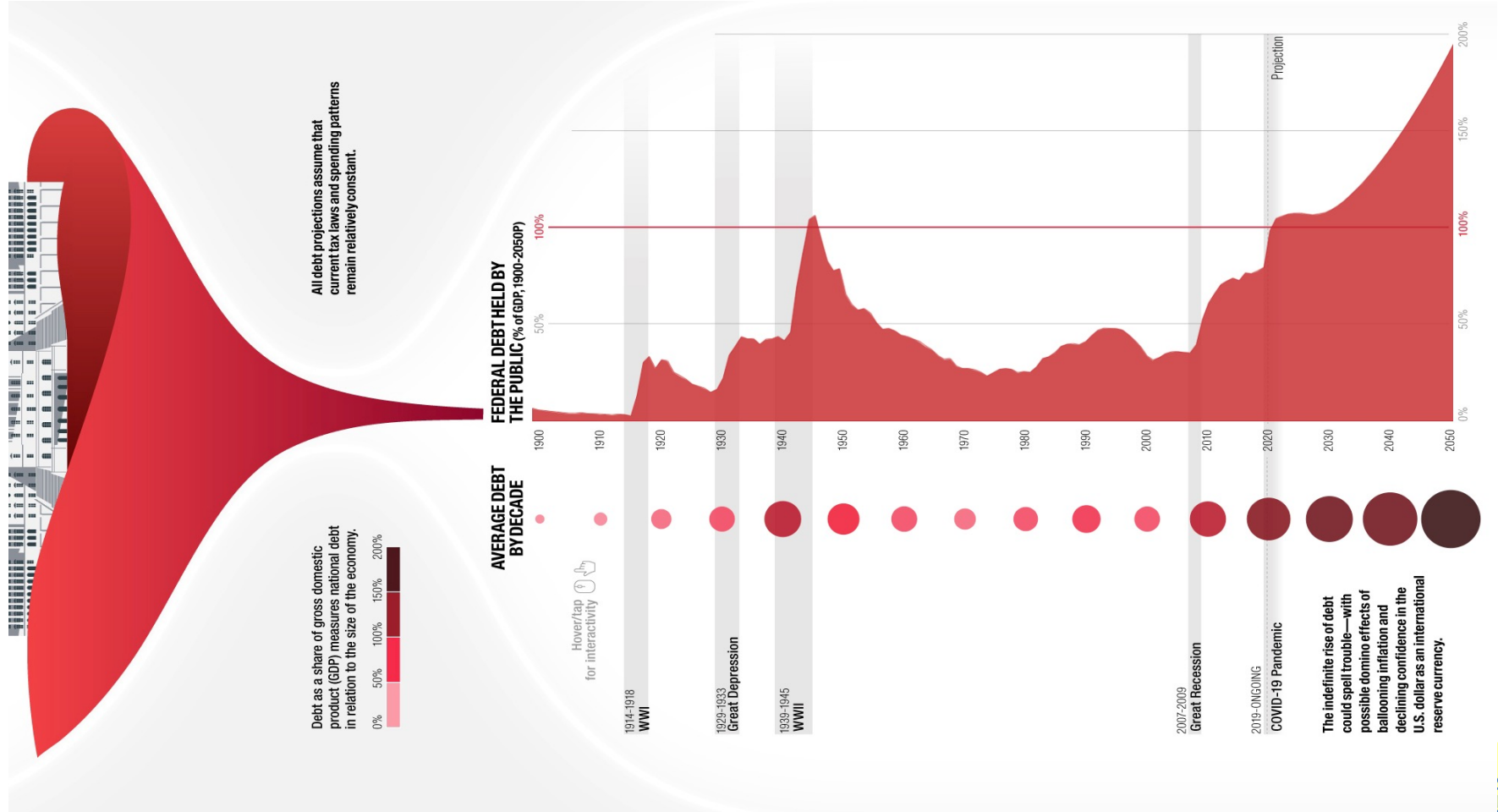
And in extreme cases, rising debt may lead to government default, resulting in economic and financial turmoil.

Can't a government that has trouble borrowing just print money to pay its bills?

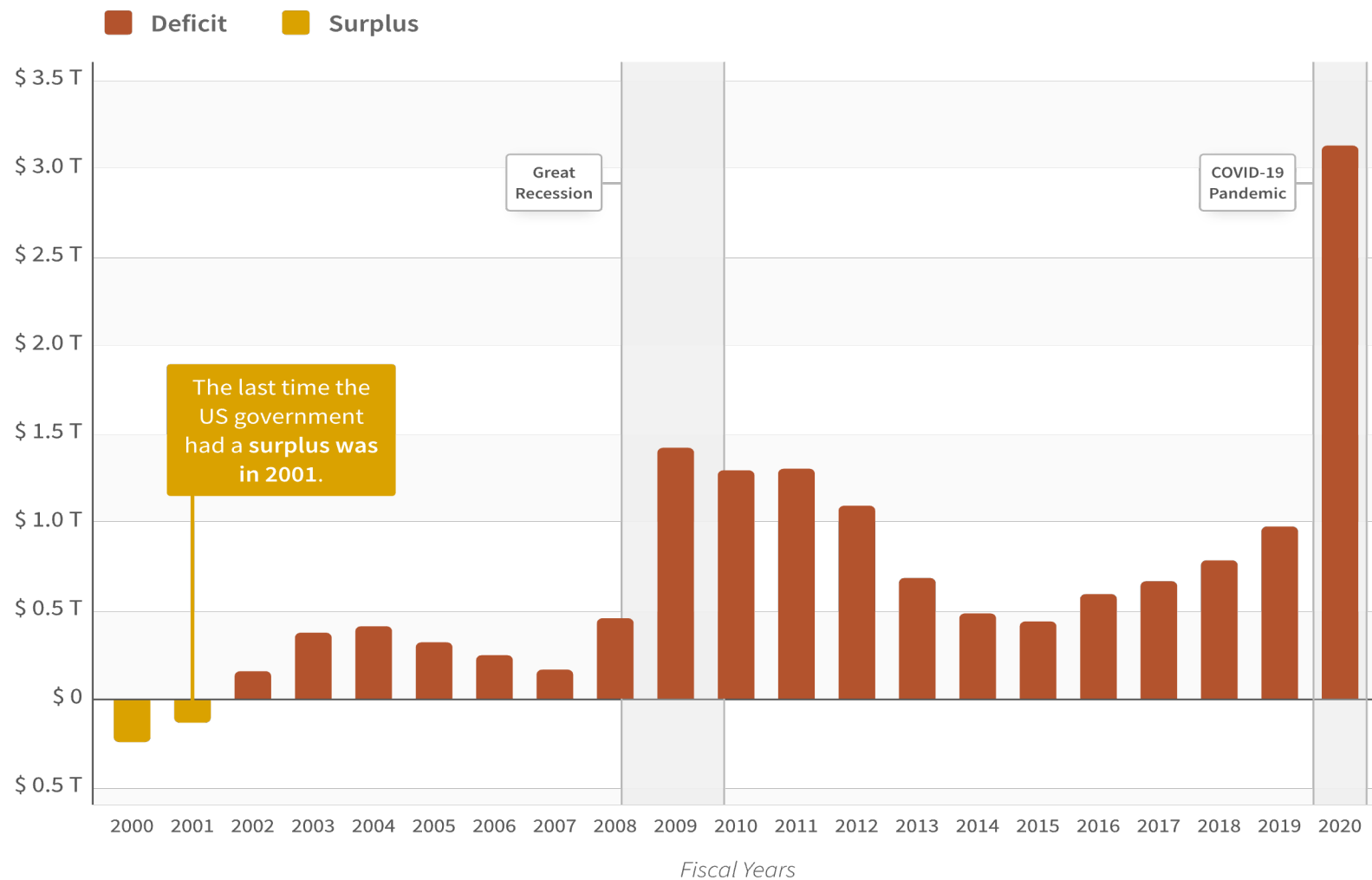
Yes, it can, but this leads to another problem: inflation. More on this in a later chapter.

DEFICITS AND DEBT IN PRACTICE

A widely used measure of fiscal health is the debt–GDP ratio.



DEFICITS AND DEBT IN PRACTICE



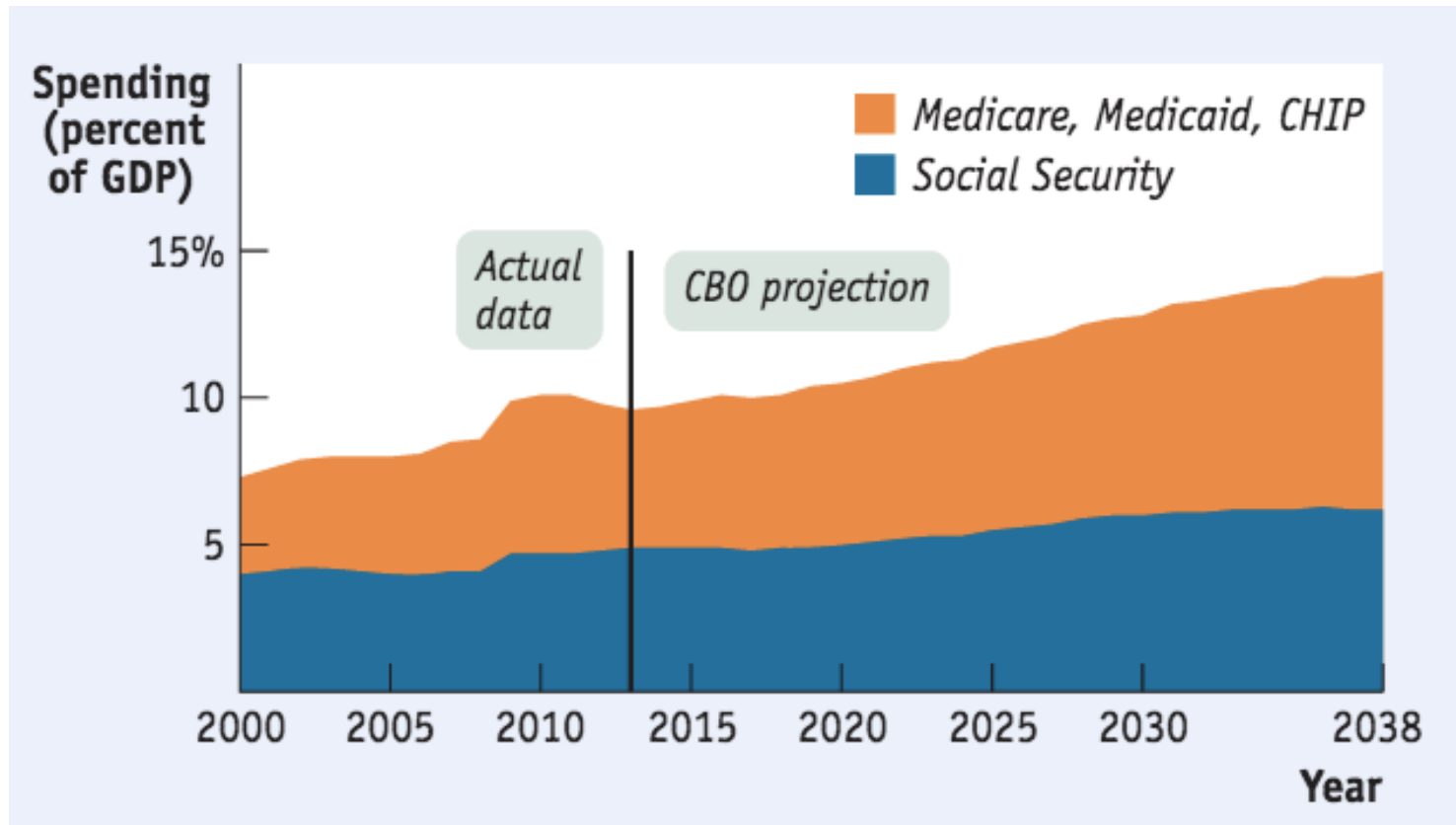
IMPLICIT LIABILITIES

Experts on long-run budget issues view the situation of the United States (and other countries such as Japan and Italy) with alarm.

Implicit liabilities: spending promises made by governments that are **effectively a debt** despite the fact that they are **not included in the usual debt statistics**.

IMPLICIT LIABILITIES

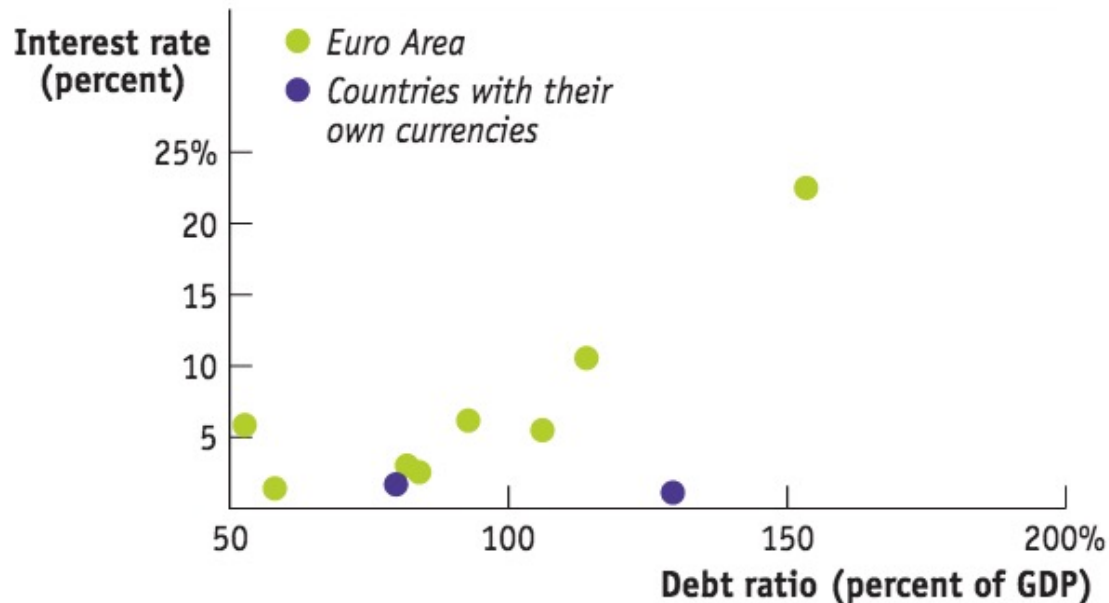
Congressional Budget Office projections:
The **aging population** plus **rising health care costs** will soon pose real problems for the federal budget.



ARE WE GREECE?

Some influential people believed the U.S. was about to head into soaring interest rates and a debt crisis like Greece experienced.

Unlike the Euro area (where countries don't have their own currency, and can't run the printing press to pay off debt if necessary) the U.S. is less scary for lenders.



Sources: Eurostat; International Monetary Fund.